

1 Properties of the Surface Covariant Derivative

1.1 Exercise 229.

Show the tensor property of the surface covariant derivative.

sol.

Let's start by recalling the definition of the surface covariant derivative.

$$\nabla_\nu T_\beta^\alpha = \frac{\partial T_\beta^\alpha}{\partial S^\nu} + \Gamma_{\nu\omega}^\alpha T_\beta^\omega - \Gamma_{\nu\beta}^\omega T_\omega^\alpha. \quad (1)$$

To show the tensor property of the given expression, we need to change all the indices with primed indices and then revert to the original indices with Jacobian matrices. The transformation law may then be expressed in terms of partial derivatives of the coordinate functions, which means

$$\nabla_{\nu'} T_{\beta'}^{\alpha'} = \left(\frac{\partial T_\beta^\alpha}{\partial S^\nu} + \Gamma_{\nu\omega}^\alpha T_\beta^\omega - \Gamma_{\nu\beta}^\omega T_\omega^\alpha \right) J_{\alpha'}^\alpha J_{\beta'}^\beta J_{\nu'}^\nu.$$

or more simply

$$\nabla_{\nu'} T_{\beta'}^{\alpha'} = (\nabla_\nu T_\beta^\alpha) J_{\alpha'}^\alpha J_{\beta'}^\beta J_{\nu'}^\nu. \quad (2)$$

By putting prime indices on 1, we get

$$\nabla_{\nu'} T_{\beta'}^{\alpha'} = \frac{\partial T_{\beta'}^{\alpha'}}{\partial S^{\nu'}} + \Gamma_{\nu'\omega'}^{\alpha'} T_{\beta'}^{\omega'} - \Gamma_{\nu'\beta'}^{\omega'} T_{\omega'}^{\alpha'}. \quad (3)$$

And now look at the terms each by each.

$$\frac{\partial T_{\beta'}^{\alpha'}}{\partial S^{\nu'}} \quad (4a)$$

$$\Gamma_{\nu'\omega'}^{\alpha'} T_{\beta'}^{\omega'} \quad (4b)$$

$$\Gamma_{\nu'\beta'}^{\omega'} T_{\omega'}^{\alpha'} \quad (4c)$$

(4a)

$$\begin{aligned} \frac{\partial T_{\beta'}^{\alpha'}}{\partial S^{\nu'}} &= \frac{\partial (T_\beta^\alpha J_{\alpha'}^\alpha J_{\beta'}^\beta)}{\partial S^{\nu'}} = \frac{\partial T_\beta^\alpha}{\partial S^{\nu'}} J_{\alpha'}^\alpha J_{\beta'}^\beta + T_\beta^\alpha \frac{\partial J_{\alpha'}^\alpha}{\partial S^{\nu'}} J_{\beta'}^\beta + T_\beta^\alpha J_{\alpha'}^\alpha \frac{\partial J_{\beta'}^\beta}{\partial S^{\nu'}} \\ &= \frac{\partial T_\beta^\alpha}{\partial S^{\nu'}} (J_{\alpha'}^\alpha J_{\beta'}^\beta J_{\nu'}^\nu) + T_\beta^\alpha J_{\alpha\omega}^{\alpha'} J_{\beta'}^\beta J_{\nu'}^\omega + T_\beta^\alpha J_{\alpha'}^{\alpha'} J_{\beta'\nu'}^\beta. \end{aligned} \quad (5)$$

Before treating 4b and 4c, let us look at the the definition of the **Christoffel symbols on embedded surfaces**.

$$\Gamma_{\beta\nu}^{\alpha} = \mathbf{S}^{\alpha} \cdot \frac{\partial \mathbf{S}_{\beta}}{\partial S^{\nu}} \quad (6)$$

(4b)

$$\begin{aligned} \Gamma_{\nu'\omega'}^{\alpha'} T_{\beta'}^{\omega'} &= \mathbf{S}^{\alpha'} \cdot \frac{\partial \mathbf{S}_{\nu'}}{\partial S^{\omega'}} T_{\beta'}^{\omega'} = \mathbf{S}^{\alpha'} \cdot \frac{\partial (\mathbf{S}_{\nu} J_{\nu'}^{\nu})}{\partial S^{\omega'}} T_{\beta}^{\omega} J_{\omega'}^{\omega'} J_{\beta'}^{\beta} \\ &= \left(\mathbf{S}^{\alpha} \cdot \frac{\partial \mathbf{S}_{\nu}}{\partial S^{\omega'}} J_{\nu'}^{\nu} + \mathbf{S}^{\alpha} \cdot \mathbf{S}_{\nu} \frac{\partial J_{\nu'}^{\nu}}{\partial S^{\omega'}} \right) T_{\beta}^{\omega} J_{\omega'}^{\omega'} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} \\ &= \left(\mathbf{S}^{\alpha} \cdot \frac{\partial \mathbf{S}_{\nu}}{\partial S^{\rho}} J_{\omega'}^{\rho} J_{\nu'}^{\nu} + \delta_{\nu}^{\alpha} J_{\nu'\omega'}^{\nu} \right) T_{\beta}^{\omega} J_{\omega'}^{\omega'} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} \\ &= (\Gamma_{\nu\rho}^{\alpha} J_{\omega'}^{\rho} J_{\nu'}^{\nu} + \delta_{\nu}^{\alpha} J_{\nu'\omega'}^{\nu}) T_{\beta}^{\omega} J_{\omega'}^{\omega'} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} \\ &= T_{\beta}^{\omega} \Gamma_{\nu\rho}^{\alpha} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} J_{\omega'}^{\omega'} J_{\nu'}^{\nu} + T_{\beta}^{\omega} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} J_{\omega'}^{\omega'} \delta_{\nu}^{\alpha} J_{\nu'\omega'}^{\nu} \\ &= T_{\beta}^{\omega} \Gamma_{\nu\rho}^{\alpha} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} \delta_{\omega}^{\rho} J_{\nu'}^{\nu} + T_{\beta}^{\omega} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} J_{\omega'}^{\omega'} J_{\nu'\omega}^{\alpha} \\ &= T_{\beta}^{\omega} \Gamma_{\nu\omega}^{\alpha} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\nu} + T_{\beta}^{\omega} J_{\alpha'}^{\alpha'} J_{\nu'\omega'}^{\alpha} J_{\beta'}^{\beta} J_{\omega'}^{\omega'} \\ &= \Gamma_{\nu\omega}^{\alpha} T_{\beta}^{\omega} \left(J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\nu} \right) + T_{\beta}^{\omega} J_{\alpha'}^{\alpha'} J_{\nu'\omega'}^{\alpha} J_{\beta'}^{\beta} J_{\omega'}^{\omega'}. \end{aligned}$$

(4c)

$$\begin{aligned} \Gamma_{\nu'\beta'}^{\omega'} T_{\omega'}^{\alpha'} &= \mathbf{S}^{\omega'} \cdot \frac{\partial \mathbf{S}_{\nu'}}{\partial S^{\beta'}} T_{\omega'}^{\alpha'} \\ &= \mathbf{S}^{\omega} \cdot \frac{\partial (\mathbf{S}_{\nu} J_{\nu'}^{\nu})}{\partial S^{\beta'}} T_{\omega}^{\alpha} J_{\alpha'}^{\alpha'} J_{\omega'}^{\omega'} J_{\rho}^{\rho} \\ &= \mathbf{S}^{\omega} \cdot \frac{\partial (\mathbf{S}_{\nu} J_{\nu'}^{\nu})}{\partial S^{\beta'}} T_{\omega}^{\alpha} J_{\alpha'}^{\alpha'} \delta_{\rho}^{\omega} \\ &= \left(\mathbf{S}^{\rho} \cdot \frac{\partial \mathbf{S}_{\nu}}{\partial S^{\beta}} J_{\beta'}^{\beta} J_{\nu'}^{\nu} + J_{\nu'\beta'}^{\rho} \right) T_{\rho}^{\alpha} J_{\alpha'}^{\alpha'} \\ &= \left(\Gamma_{\nu\beta}^{\rho} J_{\beta'}^{\beta} J_{\nu'}^{\nu} + J_{\nu'\beta'}^{\rho} \right) T_{\rho}^{\alpha} J_{\alpha'}^{\alpha'} \\ &= T_{\rho}^{\alpha} \Gamma_{\nu\beta}^{\rho} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\nu} + T_{\rho}^{\alpha} J_{\nu'\beta'}^{\rho} J_{\alpha'}^{\alpha'} \\ &= \Gamma_{\nu\beta}^{\rho} T_{\rho}^{\alpha} \left(J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\nu} \right) + T_{\rho}^{\alpha} J_{\nu'\beta'}^{\rho} J_{\alpha'}^{\alpha'}. \end{aligned}$$

Now putting all the terms in 3, we get

$$\begin{aligned}
\nabla_{\nu'} T_{\beta'}^{\alpha'} &= \mathbf{S}^{\omega'} \cdot \frac{\partial \mathbf{S}_{\nu'}}{\partial S^{\beta'}} T_{\omega'}^{\alpha'} \\
&= \frac{\partial T_{\beta}^{\alpha}}{\partial S^{\nu}} J_{\alpha}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\nu} + T_{\beta}^{\alpha} J_{\alpha\omega}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\omega} + T_{\beta}^{\alpha} \left(J_{\alpha}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\nu} \right) \\
&\quad + \Gamma_{\nu\omega}^{\alpha} T_{\beta}^{\omega} \left(J_{\alpha}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\nu} \right) + T_{\beta}^{\omega} J_{\alpha}^{\alpha'} J_{\nu'\omega'}^{\alpha} J_{\beta'}^{\beta} J_{\omega'}^{\omega} - \Gamma_{\nu\beta}^{\rho} T_{\rho}^{\alpha} \left(J_{\alpha}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\nu} \right) - T_{\rho}^{\alpha} J_{\nu'\beta'}^{\rho} J_{\alpha}^{\alpha'} \\
&= \left(\frac{\partial T_{\beta}^{\alpha}}{\partial S^{\nu}} + \Gamma_{\nu\omega}^{\alpha} T_{\beta}^{\omega} - \Gamma_{\nu\beta}^{\rho} T_{\rho}^{\alpha} \right) \left(J_{\alpha}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\nu} \right) \\
&\quad + T_{\beta}^{\alpha} J_{\alpha\omega}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\omega} + T_{\beta}^{\alpha} J_{\alpha}^{\alpha'} J_{\beta'\nu'}^{\beta} + T_{\beta}^{\omega} J_{\alpha}^{\alpha'} J_{\nu'\omega'}^{\alpha} J_{\beta'}^{\beta} J_{\omega'}^{\omega} - T_{\rho}^{\alpha} J_{\nu'\beta'}^{\rho} J_{\alpha}^{\alpha'} \\
&= (\nabla_{\nu} T_{\beta}^{\alpha}) \left(J_{\alpha}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\nu} \right) \\
&\quad + T_{\beta}^{\alpha} J_{\alpha\omega}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\omega} + T_{\beta}^{\alpha} J_{\alpha}^{\alpha'} J_{\beta'\nu'}^{\beta} + T_{\beta}^{\omega} J_{\alpha}^{\alpha'} J_{\nu'\omega'}^{\alpha} J_{\beta'}^{\beta} J_{\omega'}^{\omega} - T_{\rho}^{\alpha} J_{\nu'\beta'}^{\rho} J_{\alpha}^{\alpha'}.
\end{aligned}$$

And since $T_{\beta}^{\alpha} J_{\alpha}^{\alpha'} J_{\beta'\nu'}^{\beta}$